Queen Mary
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EBU5376
Digital Signal Processing

DSP Systems – Time-domain Responses

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Outline

- Difference Equation
  - z-transform
- Impulse Response
- Step Response

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Difference Equation - Recap

- Difference equation is often used to represent the input-output relationship in LTI systems
- However, difference equations are often **recursive**
- Solving the difference equation meaning writing the output $y[n]$ <u>independent of the past outputs</u> $y[n-1], y[n-2], \ldots$
- The main method to solve a difference equation is z-transform

Difference Equation - Recursive

$$y[n] = 0.25y[n-1], \qquad y[-1] = 1$$

n	$y[n]$
-1	1
0	0.25
1	0.25^2
2	0.25^3
3	0.25^4
…	
n	0.25^{n+1}

$$\Rightarrow \quad y[n] = 0.25^{n+1}, n \geq 0$$

We can also use z-transform to solve difference equations

Solving Difference Equation with Initial Conditions

- Steps to solve a difference equation via z-transform:
  1. Reformulate the difference equation, with the input and the output on separate sides of the equation
  2. Take the z-transform on both sides
  3. Substitute $Y[z] = Z(y[n]) = y[0] + z^{-1}y[1] + z^{-2}y[2] + \cdots$
     - Use the initial condition to simplify the equation
  4. Apply partial fraction decomposition
  5. Perform inverse z-transform to obtain the solution $y[n] = \cdots$

Difference Equation - Example

$$y[n] = 0.25y[n-1], \qquad y[-1] = 1$$

Solve the difference equation using z-transform:

Step 1: Rearrange the equation to $y[n] - 0.25y[n-1] = 0$

Step 2: The z-transform of the difference equation is given by

$$Z(y[n] - 0.25y[n-1]) = 0$$

Step 3: We have $Z(y[n]) = Y(z)$ and $Z(y[n-1]) = z^{-1}Y(z) + y[-1]$.

- *For causal signal, the unilateral z-transform $X(z) = \sum_{n=0}^{\infty} x[n]z^{-n}$ should be used, whose <u>time-shift property</u> accounts for **shifts in time** and **initial conditions**:*

$$Z(x[n-k]) = z^{-k}X(z) + \sum_{i=1}^{k} x[-i]z^{-(k-i)}$$

The term $\sum_{i=1}^{k} x[-i]z^{-(k-i)}$ accounts for the initial conditions (i.e., values of $x[n]$ for $n < 0$)

Difference Equation - Example

Step 3: We have $Z(y[n]) = Y(z)$ and $Z(y[n-1]) = z^{-1}Y(z) + y[-1]$.

The system starts producing outputs from $n = 0$. No outputs for $n < 0$.
The output sequence is $y[n] = [y[0], y[1], y[2], \dots], n \geq 0$.
No y[-1] in this sequence, because it's the initial point, not a system output.
The z-transform is

$$Z(y[n]) = y[0] + z^{-1}y[1] + z^{-2}y[2] + \cdots$$

The time-shifted sequence is $y[n-1] = [y[-1], y[0], y[1], y[2], \dots], n \geq 0$
The z-transform is

$$Z(y[n-1]) = y[-1] + z^{-1}y[0] + z^{-2}y[1] + z^{-3}y[2] + \cdots$$

Difference Equation - Example

Step 1: Rearrange the equation to $y[n] - 0.25y[n-1] = 0$

Step 2: The z-transform of the difference equation is given by

$$Z(y[n] - 0.25y[n-1]) = 0$$

Step 3: We have $Z(y[n]) = Y(z)$ and $Z(y[n-1]) = z^{-1}Y(z) + y[-1]$.

Hence the difference equation becomes

$$Y(z) - 0.25z^{-1}Y(z) - 0.25y[-1] = 0$$

$$(1 - 0.25z^{-1})Y(z) = 0.25$$

$$Y(z) = \frac{0.25}{1 - 0.25z^{-1}} = \frac{0.25z}{z - 0.25}$$

Difference Equation - Example

Step 4 : Apply the inverse z-transform to

$$Y(z) = 0.25\frac{z}{z - 0.25}$$

By checking the table for z-transform, we have

$$y[n] = 0.25 \cdot 0.25^n u[n]$$

$$= 0.25^{n+1} u[n]$$

Difference Equation - Example

- Example 2: Solve the following difference equation:

$$y[n] = 0.25y[n-1] + u[n], \qquad y[-1] = 1$$

- Solution:

Step 1: Rearrange the equation to $y[n] - 0.25y[n-1] = u[n]$

Step 2: The z-transform of the difference equation is given by

$$Z(y[n]) - 0.25Z(y[n-1]) = \frac{z}{z-1},$$

Step 3: We have $Z(y[n]) = Y(z)$ and $Z(y[n-1]) = z^{-1}Y(z) + y[-1]$.

Hence, the z-transform becomes

$$(1 - 0.25z^{-1})Y(z) = 0.25 + \frac{z}{z-1}$$

Difference Equation - Example

- Example 2: Solve the following difference equation:

$$y[n] = 0.25y[n-1] + u[n], \qquad y[-1] = 1$$

- Solution:

Step 3 (cont.): The z-transform becomes

$$(1 - 0.25z^{-1})Y(z) = 0.25 + \frac{z}{z-1}$$

$$\Rightarrow Y(z) = z\left(\frac{1.25z - 0.25}{(z-1)(z-0.25)}\right)$$

Difference Equation - Example

- Example 2: Solve the following difference equation:

$$y[n] = 0.25y[n-1] + u[n], \qquad y[-1] = 1$$

- Solution:

Step 4: Apply partial fraction to $Y(z) = z\left(\frac{\frac{5}{4}z - \frac{1}{4}}{(z-1)\left(z-\frac{1}{4}\right)}\right)$, we get:

$$Y(z) = \frac{4}{3}\left(\frac{z}{z-1}\right) - \frac{1}{12}\left(\frac{z}{z-\frac{1}{4}}\right)$$

Step 5: Apply inverse z-transform, we have $y[n] = \frac{4}{3}u[n] - \frac{1}{12} \times \left(\frac{1}{4}\right)^n u[n]$

Outline

- Difference Equation
  - z-transform
- Impulse Response
- Step Response

Impulse Response

- A LTI system can be **completely described** by its unit-impulse response $h[n]$, which is defined as the system response due to an impulse input $\delta[n]$

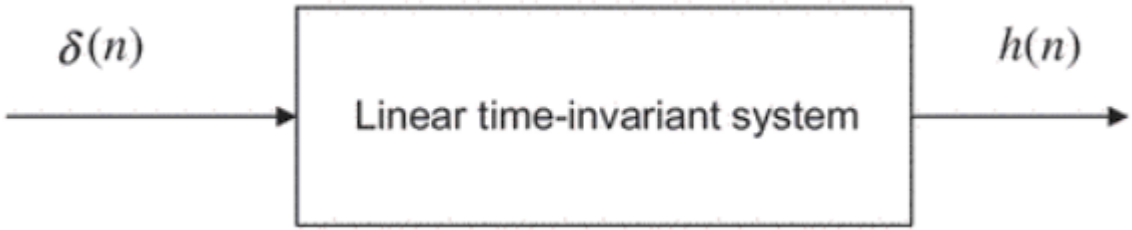


Fig: Unit-impulse response of the LTI system.

Impulse Response – System Representation

- In general, the output sequence $y[n]$ of a LTI system can be expressed as the discrete convolution of its impulse response $h[n]$ and the input sequence $x[n]$:

$$y[n] = \cdots + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + x[2]h[n-2] + \cdots$$

$$= \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$\boxed{y[n] = \textit{discrete convolution between } h[n] \textit{ and } x[n]}$$

Impulse Response

- The impulse response is only defined for LTI systems
- Other DSP systems cannot be described using the impulse response (the equation in the previous slide does not hold)
- To show this, recall that any discrete signal can be expressed as the weighted sum of shifted unit sample sequences

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

$$= \cdots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + \cdots$$

Impulse Response

- Let $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$ be the input to the LTI system
- Due to the following properties of LTI systems:
  1. Time-invariant
  2. Homogeneity
  3. Additivity
- The output of the system can be written as

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

where $h[i]$ denotes the system output when the input is $\delta[i]$

Impulse Response

- With the obtained impulse response $h[n]$, we represent the LTI system as the Fig below:

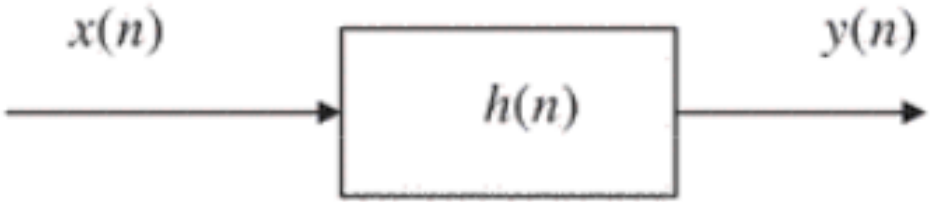


Fig.: Representation of a LTI system using the impulse response.

Impulse Response

- <u>Example</u>: Given the LTI system, $y[n] = 0.5x[n] + 0.25x[n-1]$
  1) Determine the impulse response $h[n]$
  2) Plot the impulse response $h[n]$

Impulse Response

- <u>Example</u>: Given the LTI system, $y[n] = 0.5x[n] + 0.25x[n-1]$

  1) Determine the impulse response $h[n]$

Let $x[n] = \delta[n]$, then

$$h[n] = 0.5\delta[n] + 0.25\delta[n-1]$$

The impulse response values split into three cases: $n = 0, n = 1$ and all other values of n. Hence, we have

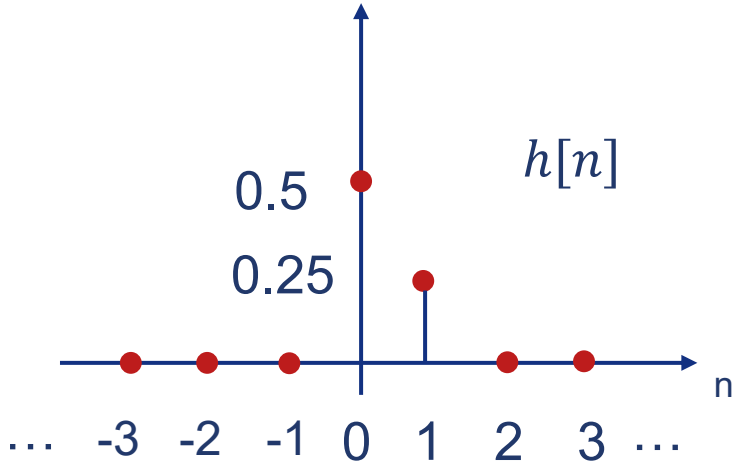
$$h[n] = \begin{cases} 0.5 & n = 0 \\ 0.25 & n = 1 \\ 0 & otherwise \end{cases}$$

Impulse Response

- Example: Given the LTI system, $y[n] = 0.5x[n] + 0.25x[n-1]$

  2) Plot the impulse response $h[n]$

$$h[n] = \begin{cases} 0.5 & n = 0 \\ 0.25 & n = 1 \\ 0 & otherwise \end{cases}$$



Impulse Response

- <u>Example</u>: Given the LTI system, $y[n] = 0.25y[n-1] + x[n]$

  1) Determine the impulse response $h[n]$

  2) Plot the impulse response $h[n]$

  (Note: The impulse response should be non-recursive. You may use the initial condition $y[-1] = 0$)

Impulse Response

- Example: Given the LTI system, $y[n] = 0.25y[n-1] + x[n]$.
  1) Determine the impulse response $h[n]$

When the input is an impulse, the system output is

$$h[n] = 0.25h[n-1] + \delta[n]$$

To obtain the non-recursive impulse response, we need to solve the above difference equation using z-transform.

An initial condition is required. For example, $y[-1] = 0$. (Will be always given in the question)

By forward and inverse z-transform, the solution to this difference equation is

$$h[n] = 0.25^n u[n]$$

Impulse Response

- Example: Given the LTI system, $y[n] = 0.25y[n-1] + x[n]$.
  1) Determine the impulse response $h[n]$

When the input is an impulse, the system output is

$$h[n] = 0.25h[n-1] + \delta[n]$$

By forward and inverse z-transform, the solution to this difference equation is

$$h[n] = 0.25^n u[n]$$

Let's verify this solution:

n	$0.25h[n-1] + \delta[n]$	$0.25^n u[n]$
0		
1		
2		
3		
4		

Impulse Response

- Example: Given the LTI system, $y[n] = 0.25y[n-1] + x[n]$.
  1) Determine the impulse response $h[n]$

When the input is an impulse, the system output is

$$h[n] = 0.25h[n-1] + \delta[n]$$

By forward and inverse z-transform, the solution to this difference equation is

$$h[n] = 0.25^n u[n]$$

Let's verify this solution:

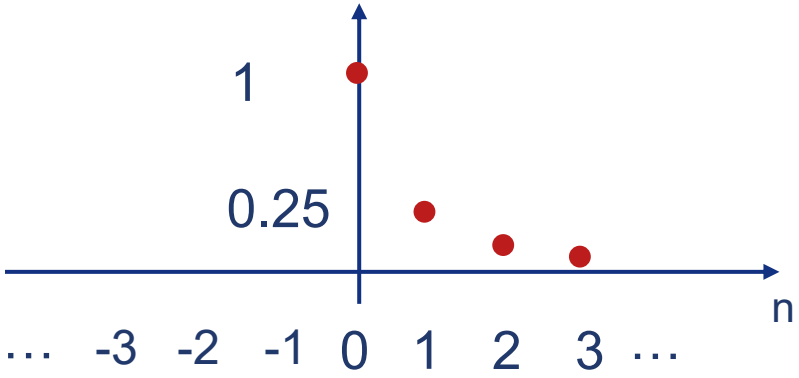
n	$0.25h[n-1] + \delta[n]$	$0.25^n u[n]$
0	1	1
1	0.25	0.25
2	0.25^2	0.25^2
3	0.25^3	0.25^3
4	0.25^4	0.25^4

Impulse Response

- Example: Given the LTI system, $y[n] = 0.25y[n-1] + x[n]$.
  2) Plot the impulse response $h[n]$

The impulse response is

$$h[n] = 0.25^n u[n]$$



Compare it to the previous impulse response, what is the main difference?

What does it imply on the system's behaviour when there is an impulse input

Summary – Impulse response

- The impulse response of a system represents its output when the input is an impulse function (a discrete-time delta function).
- It provides a system's natural response to a sudden, unit magnitude input at time zero.
- Given a recursive LTI system, the impulse response can be obtained as the solution to the difference equation, when the input is the impulse function

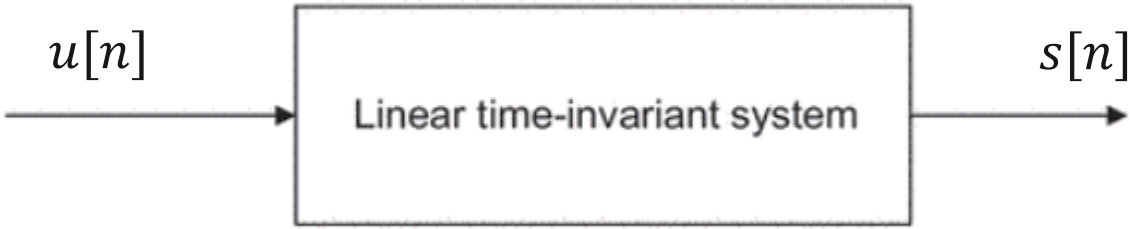
Summary – Impulse response

- Based on the impulse responses, DSP systems can be classified into two categories:
  - ✓ Finite impulse response (FIR) system
  - ✓ Infinite impulse response (IIR) system

Outline

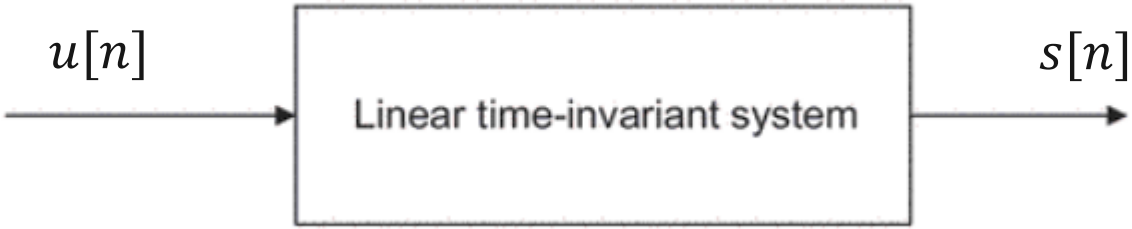
- Difference Equation
  - z-transform
- Impulse Response
- Step Response

Step Response



- Step response $s[n]$ shows how the system responds to a sudden, sustained change in input.
- For instance, when its inputs change from zero to one in a very short time.
- It is obtained as the system output, when the input is a unit step function $u[n]$

Step Response



- Similar to impulse response,
  we can obtain the step response by solving the recursive difference equation, when the input is the step function $u[n]$
- <u>Example</u>: Find the step response of the following systems:
  1. $y[n] = 0.5x[n] + 0.25x[n-1]$
  2. $y[n] = 0.25y[n-1] + x[n], \quad y[-1] = 1$

Step Response

- Example: Find the step response of the following systems:
  1. $y[n] = 0.5x[n] + 0.25x[n-1]$
  2. $y[n] = 0.25y[n-1] + x[n], \quad y[-1] = 1$

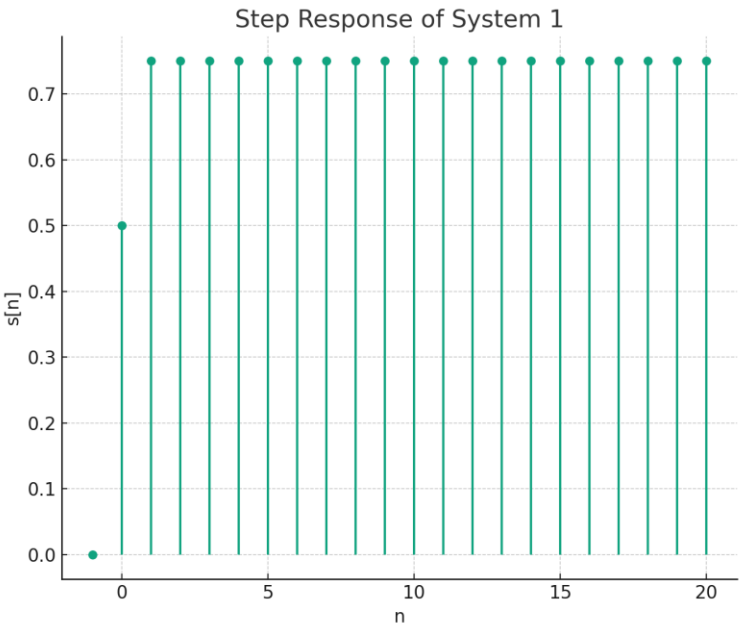
Solutions: The step responses are

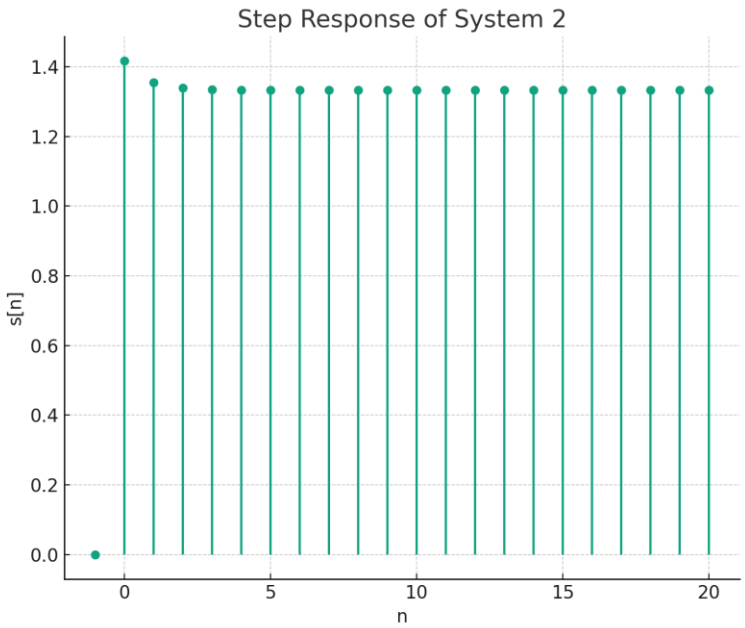
1. $s[n] = 0.5u[n] + 0.25u[n-1]$
2. $s[n] = \frac{4}{3}u[n] - \frac{1}{12} \times \left(\frac{1}{4}\right)^n u[n]$

Step Response

- Plot the step responses

By looking at these plots, can you describe the system's behaviour when there is a sudden change in the input?





Summary

- Three Key Concepts in Time-Domain Analysis:
  1. **Difference Equation**
     - Input-output relationship
  2. **Impulse Response**
     - How system reacts to a single, short burst of input
  3. **Step Response**
     - How system reacts to a sudden, sustained change

Summary

- If the difference equation is *recursive*,
  - The impulse response $h[n]$ is the solution to the difference equation, when the input is the impulse function $\delta[n]$
  - The step response $s[n]$ is the solution to the difference equation, when the input is the step function $u[n]$
- To solve the difference equation, z-transform is used

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